

How to run this project

Structure

./results

This folder contains all the results of the project's run.

./RQ1 contains the output for our RQ1. You can check the detail information in RQ1.py

./RQ2 contains the output for our RQ2. You can check the detail information in RQ2.py.

./RQ2/output need the original output data of each project. For example, if you run the project in new datasets,

the result will first be saved in **./results/rslt.save** and copy them to

./RQ2/output/new_dataatest; and the same as old.

./RQ3 contains the output for our RQ3. You can check the detail information in RQ3.py.

./rslt.plot contains the figures we generate.

./rslt.save contains the outputs of models on different datasets. You can find the full ouput in

./results/RQ2/puput

./baselineModel

This folder contains the 3 baseline models. They are OOB, PBSA, OOC.

./humla

This folder contains the package of model HumLa and ODaSC

./data

This folder contains all the datasets we use.

./data_sets contains the original datasets which do not have been pre-processed.

./data.inuse contains the datasets we finally used in our experiment.

new means the datasets extracted by our new tool

old means the datasets extracted by the old *Commit Guru*.

manul is the experimental data that we conduct manual checking tests.

./data_stream

This folder contains the function which transform the datasets into stream data for online learning models.

Usage

main.py contains the basic function to run the JIT-SDP models.

RQ1.py contains the function we use to answer our RQ1.

RQ2.py contains the function we use to answer our RQ2.

RQ3.py contains the function we use to answer our RQ3.

preprocess.py contains the function we use to pre-process the datasets.

My Experiment

1. Run RQ1.py
2. Run main.py for the models you want to check
3. Move the results of models to ./results/RQ2/output/new(old)_datasets
4. Run RQ2.py
5. Run RQ3.py
6. Check Result

Addtion

pvalue_adjustment.ipynb: This is an additional experiment in reference to the reviewers' comments. The results can be checked in **results/RQ2/performance/adjustment**

Dependency

see requirements.txt

PYTHON=3.6

pip install -r requirements.txt

Epilogue

Enjoy yourself!